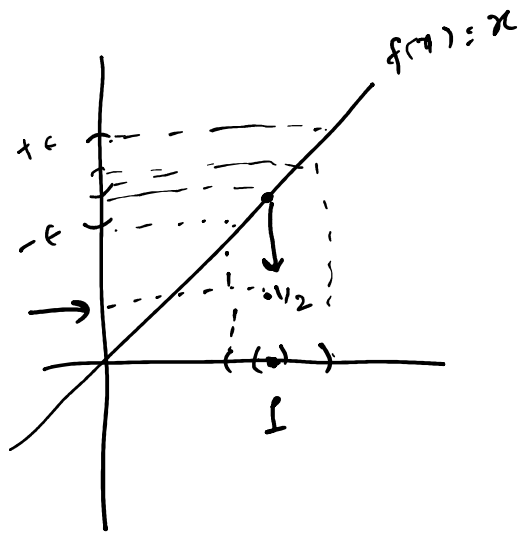
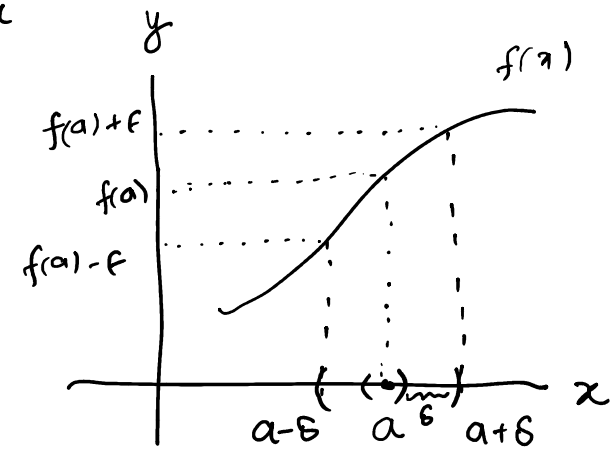
LECTURE 12

06 October 2023 07:44





Continuity: $RHL = LHL = f(a) = \lim_{x \to a} f(x)$

Eg. 1) $f(x) = \dfrac{e^{1/x}}{1 + e^{1/x}}$ $x \neq 0$ check conti. at $x = 0$

$x = 0$

$$RHL = f(0+0) = \lim_{h \to 0} f(0+h) = \lim_{h \to 0} \frac{h e^{1/h}}{1 + e^{1/h}}$$

$$= \lim_{h \to 0} \frac{h e^{1/h}}{e^{1/h}(1 + e^{-1/h})} = 0$$

$$LHL = f(0-0) = \lim_{h \to 0} f(0-h) = \lim_{h \to 0} \frac{-h e^{-1/h}}{1 + e^{-1/h}} = 0$$

$RHL = LHL = f(0) = 0$ conti. at $x = 0$

Disconti. → 4 kinds.

1) Removable disc. $RHL = LHL \neq f(a)$

Eg. $f(x) = \begin{cases} \dfrac{\sin x}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$ Check. conti $x = 0$

$\Rightarrow 1$ $x = 0$

$RHL = f(0+0) = \lim_{h \to 0} f(0+h) = \lim_{h \to 0} \dfrac{\sin h}{h} = 1$

$LHL = f(0-0) = \lim_{h \to 0} f(0-h) = \lim_{h \to 0} \dfrac{\sin(-h)}{-h} = 1$

$RHL = LHL \neq f(0) = 0$

2) Dis. of I st kind : RHL & LHL exist but $\neq$

Eg. $f(x) = [x]$

(3) Dis. of II nd kind : Neither RHL nor LHL exist.

Eg $f(x) = \sin \dfrac{1}{x}$

(4) Mixed kind disc. : one of the RHL & LHL exist but other doesn't.

Eg: $f(x) = \begin{cases} \sin \dfrac{1}{x} & x < 0 \\ 1 & x \geq 0 \end{cases}$

(1) $f(x) = \begin{cases} \dfrac{e^{1/x}}{1 + e^{1/x}} & x \neq 0 \\ 0 & x = 0 \end{cases}$ Check at $x = 0$

$RHL = f(0+0) = \lim_{h \to 0} f(0+h) = \lim_{h \to 0} \dfrac{e^{1/h}}{1 + e^{1/h}} = \lim_{h \to 0} \dfrac{\cancel{e^{1/h}}}{\cancel{e^{1/h}}(1 + e^{-1/h})}$

$$RHL = f(0+0) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} \frac{\quad}{1+e^{1/h}} = \lim_{h\to 0} \frac{\cancel{e^{1/h}}}{\cancel{e^{1/h}}(1+e^{-1/h})}$$

$$= 1$$

$$LHL = f(0-0) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} \frac{e^{-1/h}}{1+e^{-1/h}} = 0$$

Dis. of I kind at $x=0$

$LHL = f(0)$

(2) $$f(x) = \begin{cases} \dfrac{e^{1/x^2}}{1+e^{1/x^2}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$RHL = f(0+0) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} \frac{e^{1/h^2}}{1+e^{1/h^2}} = \lim_{h\to 0} \frac{1}{1+e^{-1/h^2}}$$

$$= 1$$

$$LHL = f(0-0) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} \frac{e^{1/h^2}}{1+e^{1/h^2}} = \lim_{h\to 0} \frac{1}{1+e^{-1/h^2}} = 1$$

$RHL = LHL$

Remov. desc.

$\Rightarrow$ sequential criteria.